

Fixed-Order Head Transport Forced by Annular Convergence

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July 2026

Abstract

An aggregate annular theorem would bypass root-by-root matching, but it would not bypass the underlying fixed-order moment transport. We make this necessity quantitative. A vanishing H^∞ or H^2 norm of the actual complement-to-anchor logarithmic mismatch forces every fixed coefficient to converge. Combined with the archived fixed-order total-trace and counterloop limits, this implies convergence of every fixed noisy-head moment to the finite-radius counterloop moment. The result is a necessity theorem, not an actual moving-head estimate.

1 Typed decomposition

Let $c_{\sigma,n}$ be the total noisy bulk trace, $h_{\sigma,n}$ the moment of the modulus-complete noisy head, and

$$\tau_{\sigma,n} = c_{\sigma,n} - h_{\sigma,n}.$$

Let c_n be the deterministic bulk trace, p_n the limiting pole moment, $a_n = c_n - p_n$, and $s_{k,n}$ the finite-radius counterloop moment. The archived fixed-order inputs are

$$c_{\sigma,n} \rightarrow c_n, \quad s_{k_{\sigma},n} \rightarrow p_n \quad (k_{\sigma} \rightarrow \infty)$$

for each fixed n . Define

$$g_{\sigma}(z) = \sum_{n \geq 2} \frac{\tau_{\sigma,n} - a_n}{n} z^n.$$

2 Coefficient and transport theorem

Theorem 2.1. *Fix $\rho > 0$ on which g_{σ} belongs to $X = H^\infty(\rho)$ or $H^2(\rho)$. For every fixed $n \geq 2$,*

$$|\tau_{\sigma,n} - a_n| \leq n\rho^{-n} \|g_{\sigma}\|_X.$$

Moreover,

$$\begin{aligned} |h_{\sigma,n} - s_{k_{\sigma},n}| &\leq |c_{\sigma,n} - c_n| + |s_{k_{\sigma},n} - p_n| \\ &\quad + n\rho^{-n} \|g_{\sigma}\|_X. \end{aligned}$$

Consequently, annular convergence together with the two archived fixed-order limits forces

$$h_{\sigma,n} - s_{k_{\sigma},n} \rightarrow 0$$

for every fixed order.

Proof. The coefficient of z^n in g_{σ} is $(\tau_{\sigma,n} - a_n)/n$. Cauchy's estimate proves the H^∞ case; in H^2 , any one boundary-scaled coefficient is bounded by the square norm. For the second display, rearrange the exact identity

$$\tau_{\sigma,n} - a_n = (c_{\sigma,n} - c_n) - (h_{\sigma,n} - p_n)$$

as

$$h_{\sigma,n} - s_{k_{\sigma},n} = (c_{\sigma,n} - c_n) - (\tau_{\sigma,n} - a_n) - (s_{k_{\sigma},n} - p_n)$$

and apply the triangle inequality. □

3 Protocol and boundary

The computation records the coefficient factor $n\rho^{-n}$ at $\rho = 1.41$ for orders 2, 3, 5, 8 and checks the three-term transport bound. These are scale diagnostics only; no noisy head moments are fitted.

Remark 3.1. Aggregate annular convergence can eliminate the need to pair individual roots, but it still forces aggregate head moments to follow the counterloop at every fixed order. The theorem does not prove such transport on a growing clock. Gates A–E remain false/open and no arithmetic divisor or RH claim follows.